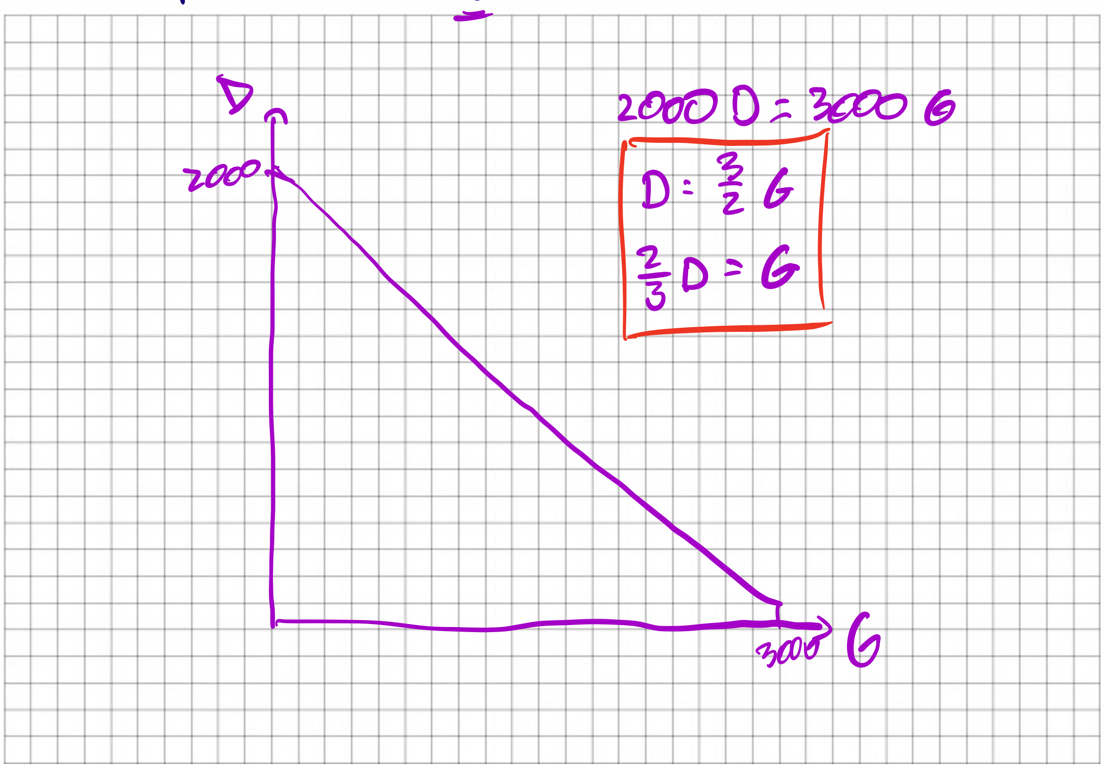
Honors Intro Micro | Homework 1 Demo

Part 1

Becca can transmogrify 2000 widgets or 3000 wigamas in one day. Set up Becaa's PPF on an x,y graph with ~~pasties~~ widgets on the vertical and ~~cakes~~ wigamas on the horizontal. What is Becca's opportunity cost of each good?

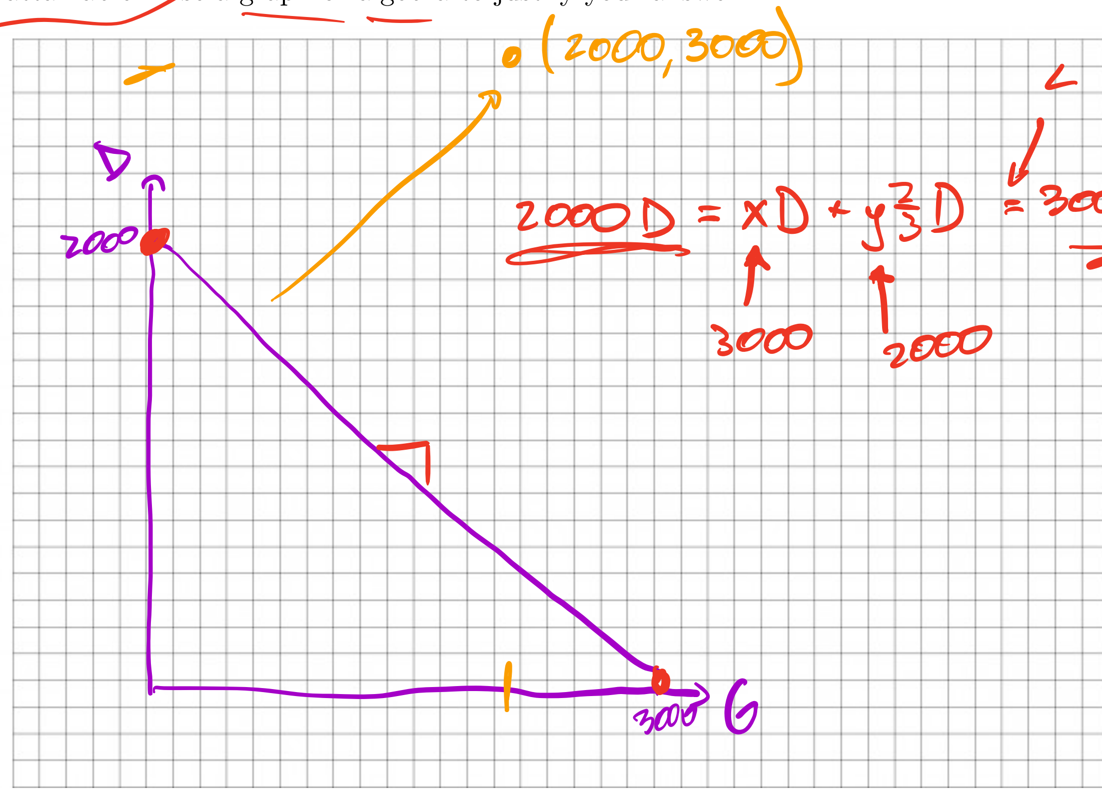


Opportunity cost of widgets: $\frac{3}{2}$ G

Opportunity cost of wigamas: $\frac{2}{3}$ D

Part 2

Suppose Becca transmogrifies 3000 widgets and 2000 wigamas in one day. Is this efficient, feasible, or unattainable? Use a graph or algebra to justify your answer.



$$\underline{2000D} = \underset{3000}{x}D + \underset{2000}{y}\tfrac{2}{3}D = \underline{3000D + \tfrac{1}{3}4000D}$$

Efficient, Feasible, Unattainable: Unatteinable

Part 3

Carole Baskine also transmogrifies widgets and wigamas, up to 10 widgets or 5 wigamas in one day. Set up a production table with both Becca and Carole's output per day. Who has the absolute advantage (AA) in widgets? Then set up an opportunity cost table with Becca and Carole's opportunity costs for each good. Who has the comparative advantage (CA) in widgets?

Production

	C	B
D	10	2000
G	5	3000

Op. Cost

	C	B
D	$\boxed{\frac{1}{2}}$	$\frac{3}{2}$
G	2	$\boxed{\frac{2}{3}}$

$10D = 5G$

$D = \frac{1}{2}G$

$2D = G$

AA in Widgets: Becca

CA in Widgets: Carole

Part 4

Suppose Becca and Ms. Baskin realize they can specialize and trade goods. After they specialize, what is a trade that would make them both better off?

Op. Cost

	C	B
D	$\boxed{\frac{1}{2}}$	$\frac{3}{2}$
G	2	$\boxed{\frac{2}{3}}$

$\frac{1}{2} < t < \frac{3}{2}$

$t = 1$

___1___ Widgets for ___1___ Wigamas

Part 5

It turns out Carole B. gets injured by a tiger, and needs to slow down her transmogrification work, and cuts her time in half. Right after, the transmogrify industry goes through a minor revolution and improves the efficiency of generating wigamas by a factor of 2. Show both changes to her PPF.